# Referee-style review

Manuscript reviewed: *An exhaustive search for four-prime Williams numbers (Carmichael numbers that are also Lucas-Carmichael) to $10^{30}$, and a verified bound of $10^{24}$ for all of them*
Review date: 1 November 2026

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| **Overall assessment** | Recommendation: major revision, with a favorable view of the core result. I found no fatal flaw in the structural proofs or the two four-prime searches. Before submission, remove the unsupported minimum claim, formalize the computational presentation, tighten the certificate, and narrow the scope. |

## What survived review

The number-theoretic core is coherent. Lemmas 2.1-2.2, Proposition 2.3, Theorem 3.1 (including both branches of the product argument and the mod-24 sharpening), Lemmas 3.5-3.6, and the polynomial-identity theorems in Appendix A re-derive correctly. The paper now attributes the pre-existing four-prime bounds and the earlier lower bounds to McIntosh and Dipra, so the main prior-art conflict in the earlier framing has been repaired.

Both four-prime certification campaigns were rerun successfully from the clean tagged checkout. The observed totals were:

| Campaign | Blocks | Pairs | Window survivors | Fermat survivors | Solutions |
|---|---|---|---|---|---|
| $n <= 10^{30}$ | 3811 / 3811 | 230,612,582,193 | 1,904 | 1 | 0 |
| $p_1 <= 500{,}000$ | 323 / 323 | 86,517,004,140 | 1,185 | 1 | 0 |

The displayed nine-prime Carmichael number was also checked: its factorization is correct, it satisfies Korselt's criterion, and exactly the four stated primes satisfy (p+1) | (n+1). The published total of 308,279,939 Carmichael numbers below $10^{24}$ is consistent with the external tabulation cited by the manuscript.

## Highest-priority revisions

**1. Remove the unsupported 'first attained at' claim in Theorem 4.1 (p. 7).**

The scan uses unordered multiprocessing and updates its stored example only when the number of matching primes strictly improves. It therefore proves that the displayed number *attains* four matches, but not that it is the least such number. Replace 'first attained at' with 'attained, for example, at' unless an ordered or explicit minimum scan is completed. This issue does not affect the $10^{24}$ nonexistence result.

**2. Make the enumeration algorithm formally executable (pp. 5-6).**

Replace the long algorithm paragraph with displayed pseudocode. Specify the campaign inputs, the exact integer definition of M*, both survivor walks, the Fermat filter, and reconstruction. Add the short argument showing why the implementation's integer-cube-root search within +/-1 must find r. Then place Proposition 3.7 immediately after the algorithm and separate mathematical completeness from implementation tests and certification.

**3. Harden and freeze the certification artifact.**

Include a block identifier on every W, FSURV, and SOLUTION record; assert the corresponding per-block counts; and explicitly assert that the Python Fermat-survivor count equals the aggregate logged count. Regenerate the reports at a clean final release tag and, ideally, archive that release with a persistent DOI. The current evidence is convincing, but these changes would make the certificate more self-contained and easier to audit.

**4. Narrow the Pomerance/Baillie-PSW claim (pp. 7-8).**

The divisibilities defining a Williams number are the backbone of Pomerance's construction, not its full definition. The construction additionally imposes congruence, Jacobi-symbol, smoothness, parity, and product-congruence conditions. A precise replacement is: *'Any counterexample produced by Pomerance's construction must exceed $10^{24}$, because that construction in particular produces a Williams number.'*

**5. Correct and moderate the four-prime heuristic (p. 8).**

Under the stated uniform-residue model the exact probability is $24/(p^2-1)$, or $24/p^2 + O(p^{-4})$, rather than exactly $24/p^2$. Replace 'confirms the prediction' with 'is consistent with the prediction.' The independence model is too speculative to justify stronger wording such as 'empty with overwhelming probability' without an explicit conditional qualifier.

**6. Qualify and document Appendix A.**

The proofs exclude polynomial families whose divisibilities hold identically in the parameter. They do not exclude families valid only on residue classes or at sporadic parameter values. Put that qualification in the abstract as well as the appendix. Explain why the exhaustive search may normalize $\gcd(a_i)=1$ by reparameterizing k, and archive frozen outputs for the stated 500/150/60 search ranges.

## Scope and organization

The strongest publishable narrative is the four-prime structure, the CRT/window algorithm, the certified campaigns, and the complete $10^{24}$ census. Appendix A.2's subset-product experiment and Appendix B's onset model require substantial methodological support but contribute little to that narrative. Move them to a supplement or separate note. The rigorous polynomial-family theorems could remain as a short appendix if their connection to the main search is stated explicitly.

### Recommended section order

- Introduction and precise comparison with McIntosh-Dipra
- Preliminaries
- Four-prime structural bounds
- Enumeration algorithm
- Completeness, validation, and certification
- Computational results
- The general $10^{24}$ census
- Discussion, open problems, and reproducibility
- Optional short polynomial-family appendix

## Claims and reproducibility language

- Replace 'every computation was re-verified' with a precise inventory of the computations independently rerun, those cross-checked against external counts, and those supported only by a single archived run.
- The $10^{24}$ scan is an independent factor-and-test pass over an external complete tabulation, not an independent tabulation of the Carmichael numbers. Call comparisons with published totals 'integrity' or 'consistency' checks.
- Clarify the sentence about Miller-Rabin square-root-of-unity splitting: particular bases can be strong liars. Say that bases are tried until a witness yields a split, with general factorization as fallback.
- Before using the new resumable $10^{24}$ rescan as evidence for a full match histogram, make worker errors fatal and assert the expected contiguous chunk set, the total 308,279,939, histogram sum equal to that total, and zero verification failures. Its present error callback can print an error and still allow a final FINISHED message. This does not affect the original zero-hit result.

- In the campaign discussion, say '1,904 and 1,185 survivor tuples, deduplicated within their respective campaigns' rather than language that might imply 3,089 globally distinct tuples.

## Editorial and layout revisions

- Shorten the title. A clearer option is: *Exhaustive searches for Williams numbers: the four-prime case and a general $10^{24}$ bound.*
- Reduce the abstract to four or five substantive sentences. It currently attempts to summarize prior history, all bounds, algorithm mechanics, the census, construction obstructions, and two heuristic programs.
- Remove the internal audit chronology: the false 3.0001 coefficient, the earlier j-design error, and 'we derived it before learning of [18]' belong in repository notes, not the journal narrative.
- Break the page-5 algorithm into a displayed procedure and the page-7 census method into short subsections such as Input data, Per-entry verification, and Integrity checks.
- Fix pagination: Appendix B ends with a dangling line on p. 11 and p. 12 contains only two references and the address. The typography is otherwise clean, with embedded fonts and no clipping.
- Add 2020 MSC classifications, keywords, and PDF title/author metadata; remove 'draft for consultation' before submission.
- Complete Pinch reference [21] with pp. 129-131 and add persistent identifiers for the central references. Replace fragile web-page-only citations where a journal paper or archived dataset is available.

## Suggested concise abstract

> *An odd squarefree composite integer is a Williams number if it is both Carmichael and Lucas-Carmichael; none is known. For a four-prime Williams number $n = p_1p_2p_3p_4$, we give a self-contained restatement and slight sharpening of bounds due to McIntosh and Dipra and derive a CRT/window enumeration. Certified computations prove that no such number satisfies $n <= 10^{30}$ or $p_1 <= 500{,}000$. Separately, a factor-and-test scan of all 308,279,939 Carmichael numbers below $10^{24}$ finds no Williams number. Code, logs, and certification records are archived.*

## Literature and publication considerations

The novelty framing is now substantially sound. The manuscript properly identifies Theorem 3.1 as a restatement of McIntosh-Dipra and presents the new contribution as the different enumeration algorithm and the stronger computational exclusions. Priority language such as 'largest bound stated in print' should nevertheless be softened to 'we know of no larger published lower bound applying without restriction on the number of prime factors.'

For a Journal of Number Theory submission, place the AI statement in a separate declaration immediately before the references and describe AI-assisted code development in Methods or Reproducibility. The declaration should name the tool, state its purposes and the author's oversight, and explicitly accept full responsibility. Venue policies differ substantially, so the destination should be selected before the wording is finalized.

### Selected sources consulted

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|---|---|
| McIntosh and Dipra (2015) | Journal of Number Theory article record |
| Pomerance (1984) | Original Baillie-PSW construction note |
| Carmichael count | OEIS A055553 |
| AI disclosure policy | Elsevier generative-AI policy for journals |